

ALEXANDER J. STERLING

Senior Full Stack Developer & Software Architect (MERN Specialist)

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PROFESSIONAL SUMMARY

Strategic and results-oriented Senior Full Stack Developer with over 10 years of professional experience in architecting, developing, and deploying high-availability, mission-critical web applications. Expert-level proficiency in the MERN ecosystem (MongoDB, Express, React, Node.js), with a specialized focus on building scalable microservices and high-concurrency event-driven systems. Proven track record of leading multi-disciplinary engineering teams to deliver award-winning SaaS and Fintech products used by millions worldwide.

Highly skilled in steering complex digital transformations, including the strategic migration of legacy monolithic architectures to modern, cloud-native infrastructures on AWS and Azure. A technical visionary committed to Engineering Excellence, Test-Driven Development (TDD), and the implementation of sophisticated CI/CD pipelines that achieve 99.9% uptime while reducing time-to-market. Recognized for optimizing application performance and improving Core Web Vitals by over 45% across enterprise-scale platforms. Passionate mentor dedicated to fostering a culture of continuous learning and technical rigor.

STRATEGIC ENGINEERING PHILOSOPHY

- **Architectural Resilience:** Prioritizing fault-tolerant designs using Circuit Breaker patterns and automated self-healing mechanisms in cloud environments.
- **Data-Centric Optimization:** Leveraging advanced database indexing, sharding, and caching strategies to ensure sub-second latency for complex analytical queries.
- **User-Centric Performance:** Designing highly responsive UIs that utilize progressive rendering, code splitting, and intelligent state management to maximize engagement.
- **Security Integration:** Deeply integrating security into the development lifecycle, ensuring adherence to SOC2, PCI-DSS, and HIPAA standards from the first line of code.
- **Scalable Leadership:** Empowering developers through transparent communication, clear technical roadmaps, and the removal of architectural bottlenecks.

COMPREHENSIVE TECHNICAL SKILLS MATRIX

Frontend & UI/UX Architecture

Core Frameworks

React (Advanced Hooks, Concurrent Mode), Next.js (SSR, ISR, SSG), Redux Toolkit, MobX.

Languages & Standards

TypeScript (Strict Mode), JavaScript (ES6-ESNext), HTML5, CSS3 (Modern Specs).

Styling & UI Systems	Tailwind CSS, Sass/SCSS, Styled Components, Material UI, Framer Motion (Animations).
Performance Tools	Webpack, Vite, Lighthouse, Chrome Profiler, Web Vitals Monitoring.

Backend & API Engineering

Runtime & Frameworks	Node.js (High Performance), Express.js, NestJS (Modular Architecture), Fastify.
API Paradigms	GraphQL (Apollo/Relay), RESTful API Design (Maturity Level 3), gRPC, WebSockets.
Processing & Messaging	RabbitMQ, Apache Kafka, Redis Pub/Sub, BullMQ (Job Queues).
Serverless Logic	AWS Lambda, Azure Functions, Google Cloud Functions.

Infrastructure, DevOps & Cloud

Cloud Ecosystems	AWS (EC2, S3, RDS, EKS, CloudFront, Route53), Microsoft Azure, Google Cloud Platform.
Containerization	Docker, Docker Compose, Kubernetes (Orchestration, Helm Charts).
Automation (CI/CD)	GitHub Actions, Jenkins, CircleCI, GitLab CI, Terraform (Infrastructure as Code).
Observability	New Relic, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana.

PROFESSIONAL WORK HISTORY

Lead Full Stack Developer | FinStream Technologies

Jan 2021 – Present | New York, NY Company Profile: A premier Fintech enterprise facilitating \$5B+ in annual cross-border transaction volume.

Leadership & Strategic Initiatives:

- ✓ **Enterprise-Scale Migration:** Orchestrated the complete re-architecture of the core payment engine from a legacy PHP monolith to a TypeScript-based microservices mesh. Implemented an API Gateway pattern to unify service access, resulting in a **300% increase in transaction capacity**.
- ✓ **Real-Time Analytics Suite:** Developed a high-frequency trading dashboard using React and Socket.io. Engineered a custom data-streaming layer that handles **20,000+ state updates per second** with negligible UI lag.
- ✓ **Financial Ledger Integrity:** Designed a double-entry bookkeeping system using MongoDB transactions and change streams, ensuring 100% data consistency across distributed nodes.
- ✓ **Global Security Compliance:** Led the engineering effort to achieve SOC2 Type II certification. Implemented automated vulnerability scanning (Snyk/SonarQube) and multi-factor authentication across all internal platforms.
- ✓ **Infrastructure Optimization:** Optimized AWS resource allocation through the use of Auto Scaling Groups and S3 Lifecycle policies, reducing monthly operational costs by **\$4,500**.
- ✓ **Team Mentorship:** Directly managed and mentored 12 developers. Established a "Peer Review Culture" that reduced production bugs by **40%** within the first year.

Senior Full Stack Engineer | GlobalCart Inc.

June 2017 – Dec 2020 | Chicago, IL Company Profile: A multi-national e-commerce giant serving 15 million active users monthly.

Technical Leadership & Architectural Design:

- ✓ **Next.js Storefront Redesign:** Led the transition of the primary storefront to Next.js. Implemented Incremental Static Regeneration (ISR) to handle 1 million+ product pages, reducing server load by **60%** while maintaining real-time price updates.
- ✓ **Design System Architect:** Founded the "Global-UI" design system using React and Tailwind CSS. Successfully integrated the library across 8 separate product domains, cutting front-end development time for new features by **35%**.
- ✓ **Search Engine Transformation:** Replaced legacy database searches with a dedicated Elasticsearch cluster. Developed custom scoring algorithms that increased search relevance and boosted overall checkout conversion by **14%**.
- ✓ **Resilient Integration Layer:** Built a robust middleware service to handle integrations with Stripe, PayPal, and Adyen. Implemented circuit breakers and exponential backoff strategies to handle third-party API outages gracefully.
- ✓ **Testing & Quality Assurance:** Standardized the testing framework using Cypress for E2E and Jest for unit testing. Achieved a consistent **85% test coverage** across core business logic services.

Full Stack Developer | CloudFlow Systems

March 2014 – May 2017 | Austin, TX Company Profile: A high-growth SaaS startup providing project management solutions for Fortune 500 agencies.

Core Development & MVP Scaling:

- ✓ **Foundational Architecture:** As one of the earliest hires, built the initial SaaS platform using the MERN stack. Designed the original database schema and the real-time notification engine.
- ✓ **High-Volume File Engine:** Developed an asset management system that supports drag-and-drop uploads for files up to 10GB. Utilized AWS S3 pre-signed URLs and Lambda-based thumbnail generation.
- ✓ **Performance Optimization:** Resolved critical memory leaks in the Node.js backend by implementing proper stream handling for large data exports, improving system stability by **95%**.
- ✓ **Collaborative Workspace:** Built a real-time multi-user editing experience using shared state and Redis, allowing users to see live updates from teammates across different time zones.

ARCHITECTURAL DEEP DIVES & CASE STUDIES

Case Study A: High-Concurrency Fraud Prevention System

Context: A payment processor required real-time analysis of incoming transactions to prevent fraudulent activities.

The Challenge: Analyzing 5,000+ data points per transaction within a 150ms window.

The Solution: Built a distributed Node.js service using the "Worker Thread" model. Used a Redis-based Bloom Filter to quickly filter known bad actors and a Kafka stream to feed data into a Python

ML model.

The Result: Successfully reduced fraudulent transactions by **22%** in the first six months, saving the client an estimated \$2.5M in chargeback fees.

Case Study B: Global Content Delivery Optimization

Context: An e-commerce platform experienced slow load times in Asia and South America.

The Challenge: Synchronizing dynamic product data across global edge locations with minimal latency.

The Solution: Implemented a multi-region AWS setup using DynamoDB Global Tables and Cloud-Front Edge Functions. Shifted heavy computations to the edge using Lambda@Edge.

The Result: Improved Time-to-First-Byte (TTFB) by **70%** for international users, directly correlating to a **9% increase in global revenue**.

SELECTED ENTERPRISE PROJECT PORTFOLIO

Project Vanguard: Secure Healthcare Portal

Stack: React, NestJS, MongoDB, Azure, WebCrypto API

- Built a HIPAA-compliant portal for patient data management and secure tele-consultations.
- Implemented client-side encryption for medical records, ensuring that sensitive data remains encrypted even within the database.
- **Impact:** Onboarded 250+ clinics and supported 1 million+ secure patient interactions.

Project Zenith: Supply Chain Intelligence Dashboard

Stack: Next.js, GraphQL, PostgreSQL, Docker, Kubernetes

- Developed a complex data visualization platform for tracking global logistics and inventory levels.
- Created interactive D3.js visualizations that allow supply chain managers to perform "What-If" analysis on logistics disruptions.
- **Impact:** Reduced inventory holding costs by **12%** through better predictive visibility.

LEADERSHIP & STRATEGIC MANAGEMENT

- **Agile Transformation:** Expert in implementing Scrum and Kanban methodologies. Facilitated transition to "Pod-based" engineering structures to increase autonomy.
- **Hiring & Scaling:** Successfully scaled the engineering department at FinStream from 15 to 60+ engineers, maintaining high quality through rigorous interview standards.
- **Technical Debt Management:** Established a "Debt-Tax" system where 20% of every sprint is dedicated to architectural cleanup and documentation.
- **Public Relations:** Frequently represented the engineering team in board meetings, stakeholder presentations, and industry conferences.

EDUCATION & ACADEMIC CREDENTIALS

Master of Science in Computer Science

Georgia Institute of Technology | 2012 – 2014

- Focus: Distributed Systems and Cloud Computing.
- GPA: 3.9/4.0.

Bachelor of Science in Software Engineering

University of Texas at Austin | 2008 – 2012

- Graduated with Honors.

PROFESSIONAL CERTIFICATIONS

AWS Architecture
Database Management
Project Management
Security

AWS Certified Solutions Architect – Professional
MongoDB Certified Developer Associate
Certified ScrumMaster (CSM)
CompTIA Security+

INTERPERSONAL & SOFT SKILLS

- **Communication:** Expert at articulating complex technical strategies to non-technical stakeholders.
- **Problem-Solving:** Exceptional ability to debug large-scale production outages and identify root causes quickly.
- **Adaptability:** Proven capacity to master new frameworks and languages within extremely short timelines.
- **Cultural Leadership:** Committed to building diverse, inclusive, and psychologically safe engineering environments.